FEYMAN

IM19

Hardware Integration Guide

Inertial Measurement Unit

Document Version: V1.4.2

Applicable Model: IM19

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Revision History

Version	Description	Date
V1.0	Initial release (sign-off: LQ)	01-2019
V1.1	Released (sign-off: LQ)	07-2020
V1.2	Released (sign-off: LQ)	01-2021
V1.3	Released (sign-off: LQ)	07-2021
V1.3.1	Released (sign-off: JL)	02-2022
V1.3.7	Released (sign-off: LQ)	10-2022
V1.4.0	Released (sign-off: LQ)	09-2023
V1.4.1	Released as IM19 Product Integration Guide (sign-off: LQ)	12-2023
V1.4.2	Hardware content moved to the current FMI document template; software protocols moved to the FMI IMU Software Integration Guide V1.3; reference-design values and SMT profile added	10-2026

NOTE This guide targets hardware engineers integrating the IM19 GNSS/INS module into a host platform. All port voltages are 3.3 V. Configuration commands and message protocols are covered in the FMI IMU Software Integration Guide V1.3. Confirm pin assignments, reference-design values and mechanical dimensions against the released schematic and mechanical drawing before production.

Contents

1 **Introduction** 4

2 **Specifications** 4

2.1 Gyroscope 4

2.2 Accelerometer 4

2.3 Attitude 4

2.4 Application Performance 4

3 **Hardware Instructions** 5

3.1 Electrical Characteristics 5

3.2 Pin Assignment 5

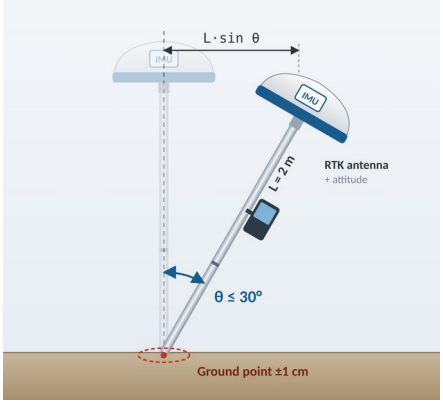
3.3 Reference Design 6

3.4 Mechanical 7

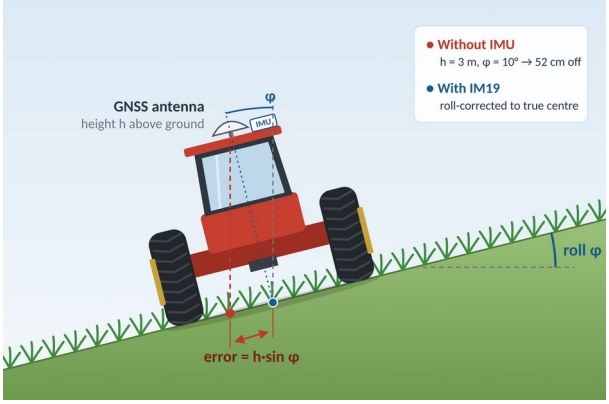
4 **SMT Precautions** 7

1 Introduction

The FMI IM19 inertial module fuses MEMS sensor data with GNSS RTK positioning data to deliver low-cost, high-precision attitude measurement, with roll and pitch accurate to within 0.05 degrees. This level of accuracy has found the IM19 widespread use in industrial applications such as tilt RTK surveys — where the RTK pole need not be held vertical, because the IM19 calculates a virtual digital level at any tilt angle — agricultural machine automation and dead reckoning.



Tilt RTK survey



Agricultural terrain compensation

2 Specifications

2.1 Gyroscope

GYROSCOPE	
Range	±1000 deg/s
Bias accuracy	±0.2 deg/s

2.2 Accelerometer

ACCELEROMETER	
Range	±8 g
Bias accuracy	±5 mg

2.3 Attitude

ACCURACY	
Heading (RTK aided)	±0.5°
Roll / Pitch (RTK aided)	±0.05°

2.4 Application Performance

APPLICATION	
Tilt survey accuracy	1 cm at up to 30° tilt
Tilt survey initialization time	1 s (95%)
Agricultural auto-steering yaw	0.5°

3 Hardware Instructions

3.1 Electrical Characteristics

Parameter	Min	Typ	Max	Units
Supply voltage	3.0	3.3	3.6	Volt
Current consumption	120	130	140	mA

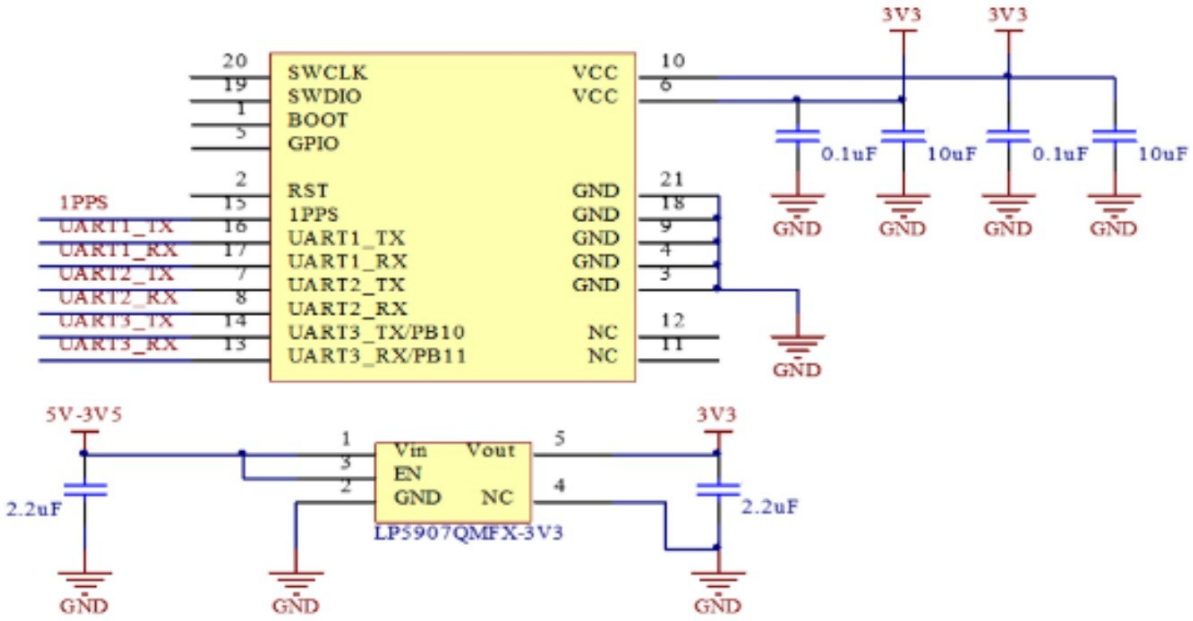
3.2 Pin Assignment

Pin	Name	Description (all port voltages are 3.3 V)
1	**BOOT**	Start modes — High: BOOT mode; NC or Low: normal working mode
2	**RESET**	Hardware reset pin (pull low 10 ms to reset module)
3	**GND**	Ground
4	**GND**	Ground
5	**GPIO**	Output IO; an active buzzer can be connected
6	**3V3**	Voltage supply
7	**UART2_TX**	UART2 output — connect to GNSS board RX
8	**UART2_RX**	UART2 input — connect to GNSS board TX
9	**GND**	Ground
10	**3V3**	Sensor power supply
11	**NC**	Reserved
12	**NC**	Reserved
13	**UART3_RX**	UART3 input — configuration, message output and firmware upgrade
14	**UART3_TX**	UART3 output — configuration, message output and firmware upgrade
15	**1PPS**	1PPS pin — connect to GNSS board 1PPS
16	**UART1_TX**	UART1 output — configuration, message output and firmware upgrade
17	**UART1_RX**	UART1 input — configuration, message output and firmware upgrade
18	**GND**	Ground
19	**SWDIO**	Debug IO (reserved)
20	**SWCLK**	Debug IO (reserved)
21	**GND**	Ground

3.3 Reference Design

UART ports 1 and 3 have the same functions; one can be connected to the main processor and the other used as a debugging interface. In addition, the BOOT pin is recommended to be routed for backup. The buzzer can be used for status indication, such as power-on and completion of installation-angle estimation.

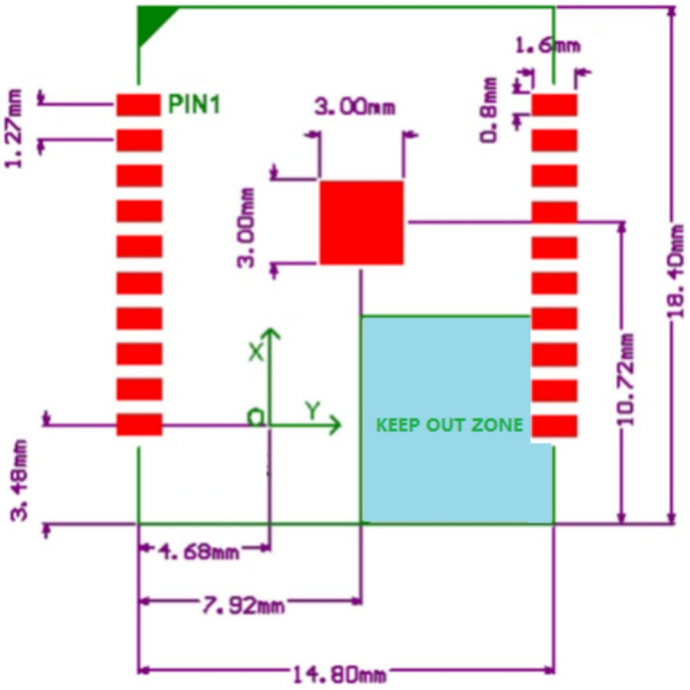
In the reference design, the 3V3 rail comes from an LP5907QMFX-3V3 LDO supplied with 3.5–5 V, with EN tied to Vin and 2.2 µF on both the input and the output. Each VCC pin (6 and 10) is decoupled with 0.1 µF and 10 µF placed close to the pin. Pins 11 and 12 are reserved and must be left unconnected.



IM19 reference design

3.4 Mechanical

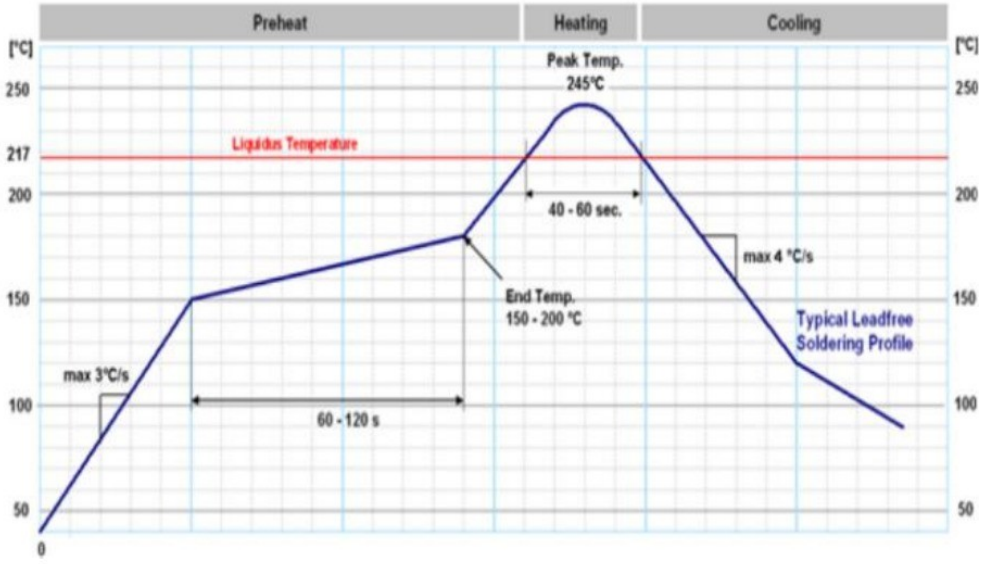
The module height is 3.2 ± 0.1 mm; the edge pad size is 1.6 mm × 0.8 mm and the middle pad size is 3 mm × 3 mm. It is recommended to place the IM19 module in a thermally stable region to avoid convection with outside air, and external stress on the module is strictly forbidden. The coordinate system is defined by the right-hand rule, as shown below.



IM19 land pattern and coordinate system (mm)

4 SMT Precautions

The IM19 modules must be baked at low temperature to remove water vapor before SMT. Bake for at least 24 hours at 60 ℃ prior to SMT. A typical lead-free reflow profile is shown below: ramp at no more than 3 ℃/s, soak for 60–120 s up to 150–200 ℃, hold above the 217 ℃ liquidus for 40–60 s with a 245 ℃ peak, and cool at no more than 4 ℃/s. For the detailed SMT temperature profile, refer to the reference document: IM19_P20 SMT Requirements.pdf.



IM19 SMT temperature curve — typical lead-free soldering profile